top_module_tb.v

```
`timescale 1ns/1ps
module top_module_tb;
reg clk;
reg rst;
reg [31:0] A;
reg [31:0] B;
reg [1:0] sel;
wire [31:0] accumulator;

top_module DUT(
.clk(clk),
.rst(rst),
.A(A),
.B(B),
.sel(sel),
.accumulator(accumulator)
);

initial begin
clk = 0;
forever #5 clk = ~clk;
end

initial begin
rst = 1;
A = 0;
B = 0;
sel = 0;
#20;
rst = 0;

sel = 2'b00;
A = 32'h3F800000;
B = 32'h40000000;
#20;

A = 32'h40400000;
B = 32'h40000000;
#20;

sel = 2'b01;
A = 32'h00003C00;
B = 32'h00004000;
#20;
```

```
A = 32'h00004200;
B = 32'h00003C00;
#20;

sel = 2'b10;
A = 32'h00003F80;
B = 32'h00004000;
#20;

A = 32'h00004040;
B = 32'h00003F80;
#20;

#100;
$stop;
end
endmodule
```